Vibhu Tummallapalli

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EDUCATION

University of North Carolina at Charlotte — Charlotte, NC
Master of Science in Artificial Intelligence — *Aug 2026 – May 2028 (Expected)*

- Focus: Applied AI/ML, MLOps & AI for Financial Services. Relevant coursework: Applied Machine Learning, Applied Artificial Intelligence, Machine Learning Operations (MLOps), Cloud Computing for Data Analysis, AI-Driven Trustworthy Software Development, Advanced Financial Modeling with AI.

East Carolina University — Greenville, NC
Bachelor of Science in Software Engineering — *May 2026*

SKILLS

Languages: Python, C#, Rust, Java, JavaScript, TypeScript, SQL

AI / ML: Agentic & multi-agent systems (CrewAI, LangChain), Model Context Protocol (MCP), Microsoft Agent Framework, LLM application development, structured prompt engineering, LLM evaluation & testing, Anthropic Claude API & Claude Code, GPT-4o & OpenAI Vision, Ollama, PyTorch

Frameworks & Web: .NET Aspire, .NET 8, ASP.NET Core, Blazor, Entity Framework Core, WPF, React, Next.js, Vite, SignalR, JWT

Cloud & Tools: GCP (App Engine, Cloud Run, Pub/Sub), Docker, GitHub Actions, GitLab CI, PostgreSQL, Bash, PowerShell

WORK EXPERIENCE

Undergraduate Research Assistant — Jun 2025 – May 2026
ECU College of Engineering and Technology — *Greenville, NC*

- Contributed to 3 faculty research projects in health tech and cybersecurity, owning prototype implementation from design through validation.
- Built data ingestion prototypes in Python, C#, and Java that processed external datasets into clean, structured outputs, informed by literature reviews and performance trade-off analyses.
- Maintained research environments (Python venvs, Maven, local LLM integrations) and managed Git repositories supporting reproducibility across the team.

PROJECT EXPERIENCE

EvalWard – LLM Prompt Regression Gate — Aug 2026
Personal Project · Bank Complaint Triage (CFPB Data) — *Charlotte, NC*

- Built a Python CLI and GitHub Actions gate that re-runs a bank complaint-triage LLM feature against a 150-case CFPB golden dataset on every prompt change, diffs it against the promoted baseline, and blocks the merge on regression.
- Scored macro-F1 across 9 stratified classes plus DeepEval relevance; set 3%/8% thresholds from a measured 2.7-point noise floor, with Dockerized runs, HTML per-case flip reports, and Teams alerts (baseline 77.3% pass, $1.60/run).

Metamorphic Testing of Multi-Agent AI Systems — Apr 2026
Research · ECU College of Engineering and Technology — *Greenville, NC*

- Designed a novel catalog of 9 metamorphic relations to test multi-agent LLM systems where traditional test oracles fail (interchangeability, redundancy invariance, graceful degradation).
- Built a reusable Python framework with multi-dimensional semantic verifiers (token overlap, difflib similarity, length ratios) and CrewAI harnesses across OpenAI and Ollama backends via LangChain, generating equivalence rates and violation reports.

MCP Tutor Agent — Jul 2026
Agentic AI Coursework · East Carolina University — *Greenville, NC*

- Built a local-first AI tutoring agent that calls external tools over the Model Context Protocol (MCP), orchestrated end to end with .NET Aspire.
- Implemented two MCP servers in C# (stdio and streamable HTTP transports) and wired their tool catalogs into an LLM agent using the Microsoft Agent Framework.
- Streamed responses token-by-token from an ASP.NET Core minimal API to a Blazor Fluent UI chat, running fully on local models via Ollama.

Dental Charting Tool — Jun 2025 – Jan 2026
Applied Research · ECU School of Dental Medicine — *Greenville, NC*

- Built a WPF (.NET)/C# desktop app integrating GPT-4o Vision for radiographic analysis, FDI tooth-numbering charting, CDT-coded procedure cataloging, and urgency-based appointment scheduling.
- Engineered a 5-step prompt pipeline (observation, chart cross-reference, medical alerts, CDT recommendations, urgency triage) for clinically contextualized output; presented at ECU RCAW and CORAS 2026.

NC Early Mathematics Placement Testing — Aug 2025 – May 2026
Online Testing System · Senior Capstone — *Greenville, NC*

- Built a full-stack web platform (React 19/Vite, .NET 8, EF Core, PostgreSQL) used by 60 NC high schools and 5,000+ students, with a version-controlled question bank, automated grading, and role-based access (JWT + 2FA).
- Added real-time SignalR proctor monitoring and an async email pipeline (GCP Pub/Sub + Python Cloud Function); deployed to GCP through GitHub Actions CI/CD in a 5-member Agile team.